



Вычислительная геометрия и алгоритмы компьютерной графики

Лекция №9.2: Наложение текстур

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===== vertex shader =====

```
in vec4 a_position;
in vec2 a_texCoord;
out vec2 v_texCoord;
void main()
{
    v_texCoord = a_texCoord;
    gl_Position = a_position;
}
```

===== fragment shader =====

```
uniform sampler2D u_map;
in vec2 v_texCoord;
layout (position = 0) out vec4 u_color;
void main()
{
    o_color = texture(u_map, v_texCoord);
}
```

===== init =====

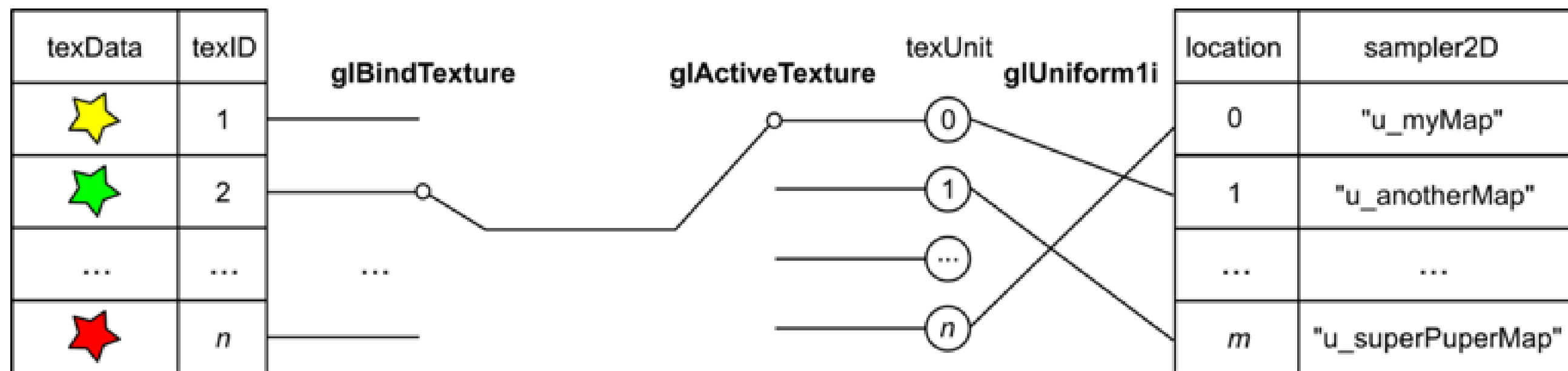
```
GLsizei texW, texH;  
GLvoid *image = loadTexture(filename, &texW, &texH);  
GLuint texID;  
glGenTextures(1, &texID);  
glBindTexture(GL_TEXTURE_2D, texID);  
glTexImage2D(GL_TEXTURE_2D, 0, GL_RGBA, texW, texH, 0, GL_RGBA,  
             GL_UNSIGNED_BYTE, image);  
delete[] image;  
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_WRAP_S, GL_CLAMP_TO_EDGE);  
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_WRAP_T, GL_CLAMP_TO_EDGE);  
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_MIN_FILTER, GL_NEAREST);  
glTexParameteri(GL_TEXTURE_2D, GL_TEXTURE_MAG_FILTER, GL_NEAREST);  
GLint mapLocation = glGetUniformLocation(program, "u_map");
```

===== draw =====

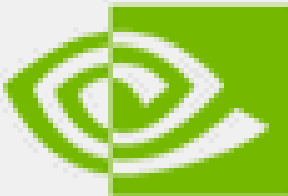
```
glActiveTexture(GL_TEXTURE0 + texUnit);  
glBindTexture(GL_TEXTURE_2D, texID);  
glUniform1i(mapLocation, texUnit);
```

===== cleanup =====

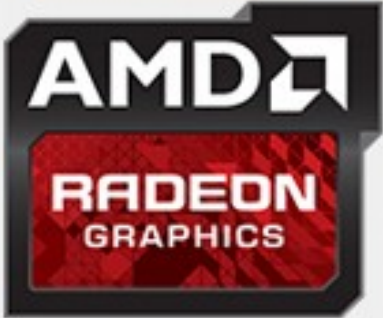
```
glDeleteTextures(&texID);
```



Texture Mapping Unit (TMU)

Name	NVIDIA GeForce RTX 5060			Lookup
GPU	GB206	Revision	A1	 NVIDIA
Technology	5 nm	Die Size	181 mm ²	
Release Date	May 19, 2025	Transistors	21900M	
BIOS Version	98.06.21.40.E6			 ☒ UEFI
Subvendor	Zotac/PC Partner	Device ID	10DE 2D05 - 19DA 1780	

ROPs/TMUs	48 / 120
Shaders	3840 Unified

Name	AMD Radeon(TM) 890M Graphics			Lookup
GPU	Strix Point	Revision	C1	
Technology	4 nm	Die Size	233 mm ²	
Release Date	Jun 3, 2024	Transistors	Unknown	
BIOS Version	023.010.001.019			 ☒ UEFI
Subvendor	AMD/ATI	Device ID	1002 150E - 1002 0124	
ROPs/TMUs	16 / 64	Bus Interface	PCIe x16 4.0 @ x16 4.0 ?	
Shaders	1024 Unified	DirectX Support	12 (12_2)	